

CS263 Midterm Project Report

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Abstract

Large vision-language models (LVLMs) suffer from hallucinations, generating plausible text that is not supported by the visual input. Visual Contrastive Decoding (VCD) is proposed to mitigate hallucination by contrasting the model’s output distribution on the original image with that on a globally distorted version. However, the distortion is query-agnostic: the same Gaussian-noise corruption is applied regardless of what the question asks. Recent work attempts to inject awareness into the auxiliary view either through salience-based masking or through latent-space interventions. Self-Augmented VCD introduces genuine query-conditioning by having the LVLM select among predefined augmentations. We push this direction further by treating the LVLM as a query-conditional corruption planner. Then we propose Query-Aware Contrastive Decoding (QACD), which combines planning with attention-based grounding. QACD is training-free, uses no external models, and adds only a single planner forward pass per query.

1 Introduction

Large vision-language models (LVLMs) such as LLaVA (Liu et al., 2023) and Qwen-VL (Bai et al., 2023) have achieved remarkable performance on a wide range of multimodal tasks. However, they continue to suffer from a fundamental reliability problem, hallucination, where the generated text describes objects, attributes, or relations that are not actually present in the visual input (Bai et al., 2024). This issue limits their use in sensitive domains and erodes trust in their outputs. Mitigation strategies broadly fall into two categories. Training-based methods, including instruction tuning, reinforcement learning from human feedback, which can be effective but are expensive, model-specific, and not transferable. Inference-time methods, in contrast, intervene only during decoding and can

be applied as plug-ins to any pretrained LVLM. Among these, Visual Contrastive Decoding (VCD) (Leng et al., 2024) is a particularly simple and effective approach: it contrasts the LVLM’s output distribution on the original image against that on a Gaussian-noise-distorted version. The distorted distribution captures hallucination-prone behavior driven by language priors when visual evidence is degraded; subtracting it from the original suppresses those priors.

A key limitation of VCD, however, is that its construction is query-agnostic: the same global Gaussian noise is applied regardless of what the question asks. Subsequent works have tried to inject awareness, but most do so in ways that remain decoupled from the explicit query. Mask What Matters (Chen et al., 2026), for instance, masks the most salient region identified via ViT attention, a procedure that happens to align with POPE-style object questions, where salient objects are typically the queried ones, but breaks down for attribute questions, peripheral objects, or open-ended generation where salience and query diverge. Self-Augmented Visual Contrastive Decoding (SA-VCD) (Im et al., 2026) takes a step closer to genuine query-conditioning by having the LVLM select among six predefined augmentations, but the chosen operation is still applied globally with fixed parameters.

In this work, we push this direction further by combining symbolic planning with attention-based grounding. The LVLM is prompted, in a single forward pass, to produce a structured corruption recipe specifying which operation to apply and how strongly; and the region is extracted directly from the model’s cross-attention on the query during the same forward pass. This division of labor plays to each component’s strengths: the language model is good at semantic reasoning about which operations are disruptive, while attention is more reliable than verbalized spatial language for localizing query-relevant regions. The prompt uses an adversarial

framing inspired by GAN-style setups: the model proposes an edit that would mislead another vision-language model attempting to answer the query, but we emphasize this is a prompt-engineering device, not actual adversarial training.

A second contribution concerns which corruptions are appropriate. We observe that VCD’s effectiveness depends on whether the amateur distribution is anti-correlated with the ground-truth answer. A corruption is harmful when its output systematically resembles a plausible ground truth, the canonical case being zero-masking on object-existence queries, where the masked region creates “absence” evidence that matches the correct answer for negative samples, causing VCD to subtract correct signal. Conversely, seemingly aggressive operations like color inversion are safe: they push the model toward a different specific answer that does not resemble any ground-truth value. We articulate this principle and restrict our operation set accordingly.

2 Related Work

2.1 Inference-time hallucination mitigation.

Hallucination in LVLMs is well-documented (Bai et al., 2024) and evaluated by benchmarks including POPE (Li et al., 2023), MMVP (Tong et al., 2024), and HallusionBench (Guan et al., 2024). Inference-time methods avoid the cost of retraining and broadly divide into contrastive decoding, attention/activation intervention, and external verification approaches.

2.2 Contrastive decoding.

VCD (Leng et al., 2024) established the framework by contrasting outputs on original vs. globally noised images. MARINE (Zhao et al., 2024) applies classifier-free guidance from external object detectors. Subsequent work explores alternative ways of constructing the amateur branch: Lee et al. (Lee et al., 2024) empirically compare down-sampling and InstructPix2Pix-based editing; Mask What Matters (Chen et al., 2026) uses ViT-attention saliency masking. These approaches modify what the amateur branch sees but do so independently of the text query.

2.3 Latent-space and attention-based methods.

A parallel line of work intervenes inside the model rather than on the image. VTI (Liu et al., 2024)

steers latent representations toward hallucination-robust directions. ASD (Su et al., 2025) performs bidirectional hidden-state intervention. MaskCD (Deng and Yang, 2025) masks vision-sensitive attention heads. ASCD (Wang et al., 2025) modulates attention scores during contrastive decoding. These methods are powerful but operate on implicit signals, attention shifts, hidden-state directions, that are harder to inspect than image-space operations.

2.4 Query-aware methods.

Closest to our work, SA-VCD (Im et al., 2026) introduces query-conditional *selection* among predefined augmentations via meta-prompting. IAVA (Li et al., 2025) identifies query-irrelevant tokens by comparing attention under two instructions. Our contribution extends this direction along two axes: (i) we combine symbolic planning of which operation to apply with attention-derived localization of where to apply it and (ii) we explicitly articulate evidence anti-correlation as the operation selection principle, which prior methods follow only implicitly.

3 Method

3.1 Preliminaries

Visual Contrastive Decoding. Given an image v and text query x , an LVLM parameterized by θ generates a response autoregressively by sampling from $p_\theta(y_t | v, x, y_{<t})$. Visual Contrastive Decoding (VCD) (Leng et al., 2024) constructs a distorted image v' (typically by adding Gaussian noise) and computes a contrastive distribution. Let $\ell_v = \ell_\theta(y_t | v, x, y_{<t})$ and $\ell_{v'} = \ell_\theta(y_t | v', x, y_{<t})$ denote the logits under the original and distorted images. The VCD distribution is:

$$p_{\text{VCD}} = \text{softmax}[(1+\alpha)\ell_v - \alpha\ell_{v'}], \quad (1)$$

where α controls the contrastive strength. The amateur distribution $\ell_{v'}$ captures hallucination-prone behavior driven by language priors when visual evidence is degraded; subtracting it from the original suppresses those priors.

3.2 Choosing Corruptions for VCD

VCD subtracts the amateur logits from the expert logits. For this to suppress hallucination rather than correct answers, the amateur distribution must be anti-correlated with the ground truth, the corrupted

image should push the model away from the correct answer, not toward it.

This admits two safe categories of corruption. The first is **content-neutral degraders** (e.g., Gaussian blur, downsampling, additive noise): they increase uncertainty without committing to any specific answer. The second is **evidence flippers that cannot reproduce ground truth** (e.g., color inversion): they push the model toward a different specific answer (“cyan” instead of “red”) that does not resemble any plausible ground-truth value.

The principle rules out corruptions whose output systematically resembles a plausible ground-truth label. The canonical case is zero-masking on object-existence queries: the masked region creates strong “absence” evidence, which is the correct answer for negative samples. When the amateur reinforces this correct answer, VCD subtracts correct signal. We restrict our operation set accordingly.

3.3 The QACD pipeline

QACD consists of three stages:

Stage 1: Recipe planning. The LVLM is prompted with the image v , the query x , and a meta-prompt instructing it to act as an adversary proposing an edit that would mislead another vision-language model. The prompt elicits a parseable output specifying a target concept, an operation, a region, and an intensity:

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TARGET: <one-line description>
OPERATION: <op>
INTENSITY: <1, 2, or 3>
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The planner runs in a single forward pass. On parse failure, we fall back to applying noise to the center region of the image, preserving the structured-recipe form of the method.

Stage 2: Attention-derived region. The corruption region is not verbalized by the planner (LVLMs are weak at precise spatial language) but extracted from the cross-attention computed during the planner’s forward pass. Concretely, we collect the attention weights from the TARGET tokens to the image patches at a mid-layer of the LLM, average across attention heads, and threshold the resulting heatmap to obtain a binary mask of query-relevant image regions. This grounds the localization in the model’s own assessment of what is relevant, rather than relying on its verbal description of regions.

Stage 3: Recipe execution and contrastive decoding. The planned operation is applied to the attention-derived region of the original image as a lightweight pixel-space transformation

Op	Mechanism	Type
blur	Gaussian blur	Degrader
downsample	Heavy pixelation	Degrader
noise	Gaussian noise	Degrader
obscure	Blur + darken	Degrader
r-noise	Noise in region	Degrader
desat	Collapse to grayscale	Degrader
invert	Color/channel inversion	Flipper

Table 1: Operations in our recipe space.

(sub-millisecond per image, no learned models involved). Standard VCD-style contrastive decoding (Eq. 1) is then performed with the corrupted image as the amateur branch.

3.4 Operation set

Following the design principle, we curate an operation set that is either content-neutral (degraders) or anti-correlated with any plausible ground-truth answer (evidence flippers that cannot reproduce the original). The seven operations are listed in Table 1.

We deliberately exclude operations that risk ground-truth alignment, most notably zero-masking and mean-fill, which produce strong “absence” evidence and would harm performance on negative POPE samples where the queried object is genuinely absent.

The planner produces a recipe of (operation, intensity), giving $7 \times 3 = 21$ possible combinations. The region of application is derived separately from the planner’s cross-attention.

4 Experiment Setup

4.1 Model

We use LLaVA-1.5-7B as our primary LVLM, both as the corruption planner and as the target model under contrastive decoding. If time permits, we additionally evaluate Qwen-VL.

4.2 Benchmark

Our primary benchmark is POPE (Li et al., 2023), which evaluates object hallucination via yes/no questions about object existence. POPE includes three sampling strategies (random, popular, adversarial) across the MSCOCO, A-OKVQA, and GQA datasets. We report accuracy, precision, recall, and F1. If time permits, we additionally evaluate on MMVP (Tong et al., 2024) and HallusionBench (Guan et al., 2024).

4.3 Baseline

1. Greedy decoding
2. Vanilla VCD (Leng et al., 2024)
3. SAVCD (Im et al., 2026)

5 Next Step

Experiment conduction and result analysis

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